

Arhaan Penwala

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EDUCATION

Vellore Institute of Technology

Chennai, India

B.Tech Computer Science and Engineering (AI and Robotics), CGPA: **9.13/10.0**

Expected May 2027

- Relevant Coursework: Data Structures and Algorithms, Operating Systems, DBMS, Computer Networks, OOP, Compiler Design

EXPERIENCE

Co-Founder and CTO

Nov '24 – Present

V Help | Flutter, Supabase (PostgreSQL, Deno Edge Functions), Firebase, Azure

VIT Chennai

- Own the Flutter client and Supabase backend of a live campus super-app serving **1,000+ registered students**, as top contributor of 3 core engineers
- Closed a privilege-escalation surface by relocating **24** client write paths into server-authorized Deno Edge Functions backed by PostgreSQL Row Level Security
- Migrated **500+ accounts** from two separate Firebase projects onto a single Supabase identity via a custom JWT exchange, at **zero re-registration**
- Cut release turnaround from **45 minutes** of manual work to **6 minutes** by automating build and distribution through GitHub Actions CI/CD

Software Development Engineer Intern

Dec '25 – Feb '26

BPO Integra India Private Limited

Remote

- Shipped a reusable **React** component library and Node.js/Express REST services across **6** pages of a live production site, end to end from UI to database
- Cut response times **23%** across **4** production REST endpoints by collapsing N+1 query patterns and adding composite MySQL indexes

AI Research Intern

May – Jun '25

The Indian Hotels Company Limited (Taj Hotels, Tata Group)

Mumbai, India

- Built a Python cost model over **5** enterprise LLM vendors and **15** hospitality use cases, projecting a **35%** spend reduction that drove leadership's vendor selection

PROJECTS

PolarisGCS | FastAPI, PostgreSQL/TimescaleDB, Redis, React/TypeScript, Docker, Kotlin

Feb '26 – Present

- Built an operator dashboard in **React, TypeScript and Vite** rendering live per-drone markers on a Leaflet map, HLS video via hls.js, and separate operator and admin route trees behind Firebase Auth
- Served **15 concurrent MAVLink vehicles** to the browser over REST and WebSocket APIs on stateless FastAPI instances, fanning out through Redis pub/sub with automatic client reconnect
- Persisted time-series telemetry in TimescaleDB with Alembic migrations and continuous aggregates, and containerized the full stack into a **single-command** Docker Compose deployment

PawGuard | Flutter, Riverpod, Supabase (PostgreSQL, Deno Edge Functions), Solidity, Razorpay

Jun '25 – Present

- Built a cross-platform pet adoption and rescue platform across **6** feature modules including marketplace, crowdfunding and medical records, contributing **149 of 177** commits
- Bridged Firebase Auth to PostgreSQL by passing ID tokens into **Row Level Security** policies on every table, eliminating a duplicate user store entirely
- Recorded pet ownership and donation transparency on **Polygon** via 3 Solidity contracts, writing on-chain only through a server-held wallet so clients never sign transactions
- Shipped **17** Deno Edge Functions verifying Firebase JWTs against Google JWKS, keeping payment and wallet secrets out of the client binary
- Integrated Razorpay checkout with server-side HMAC verification and **SECURITY DEFINER** PostgreSQL RPCs for atomic order placement and stock decrement

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, Java, C/C++, SQL, Dart, HTML/CSS, Kotlin

Frontend: React.js, Vite, Flutter, Riverpod, Material Design, Responsive Design, Leaflet, PySide6

Backend: Node.js, Express.js, FastAPI, REST APIs, WebSockets, Redis Pub/Sub, JWT Authentication, Deno Edge Functions

Databases: PostgreSQL/TimescaleDB, MySQL, MongoDB, Redis, SQLite, Supabase, Firebase

Infrastructure & Concepts: Docker, AWS, Azure, Git, GitHub Actions, CI/CD, Postman, REST API Design, Authentication and Authorization, Data Structures and Algorithms, OOP

ACHIEVEMENTS AND PUBLICATIONS

2nd Place out of **500+ teams**, Hack N Droid '25 | **Publication:** gait dataset, IEEE DataPort, DOI 10.21227/ng5-b637